

Title

Outcomes of Onsite Telemedicine-Facilitated Hepatitis C Treatment Integrated Within Opioid Treatment Programs: Effectiveness and Patient Satisfaction Results

Abstract

Treatment barriers reduce hepatitis C cure rates in opioid-using populations. We conducted a hybrid effectiveness-implementation analysis comparing telemedicine provision within opioid treatment programs (OTPs) versus standard referral to offsite hepatology. The integration approach involved on-site clinical coordinators, telemedicine consultation infrastructure, and care coordination. Among participants receiving telemedicine within their OTP, hepatitis C viral clearance was achieved in 81% (95% CI 76-85%) compared with 64% (95% CI 58-70%) among the referral group. Treatment completion rates were 89% in the integrated model versus 71% in the referral model. Patient satisfaction with healthcare provider contact duration was significantly higher in the telemedicine group (mean 7.4 vs. 5.1 on 10-point scale, $p < 0.001$). Organizational factors associated with superior effectiveness included on-site care coordination staff consistency, manageable clinical provider-to-patient ratios, and proactive participant identification. Telemedicine-enabled co-location of hepatitis C services with substance use treatment substantially improved both virologic outcomes and participant satisfaction.